

Stability Analysis of the CV and CC Control Loops

2-Channel Pre-Regulated Power Supply — study loops (simple loops.kicad_sch, sheets V_loop and I_loop)

Companion document to V_loop_transfer_functions.ipynb and I_loop_transfer_functions.ipynb — 2026-07-06

1 Overview and method

Both regulator loops share the same four-block structure (Figure 1): an op-amp **error amplifier** with a frequency-shaping feedback network (the compensator), a **driver + pass stage** (small NPN driving a TIP125 PNP Darlington), an **output network**, and a **sense amplifier** that scales the regulated quantity back to the error amp.

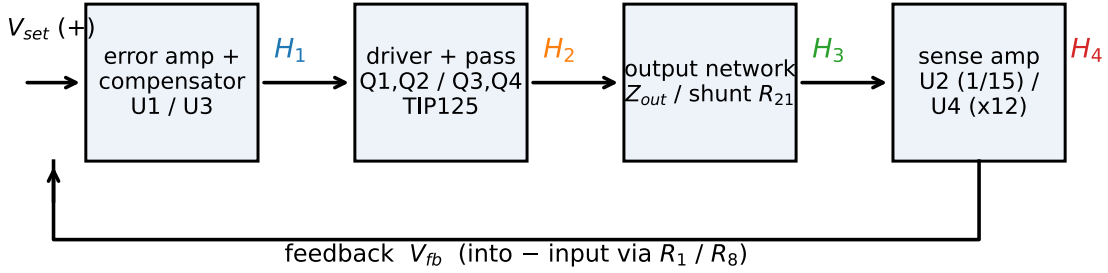


Figure 1: Common loop structure. The voltage loop senses V_{out} (U2, gain 1/15); the current loop senses the shunt voltage $R_{21}I$ (U4, gain 12).

Each block is described by its small-signal transfer function $H_1 \dots H_4$; the **loop gain** is

$$T(s) = H_1(s) H_2(s) H_3(s) H_4(s).$$

The error amp is inverting for the fed-back signal, which supplies the 180° that makes the feedback negative; stability is then read from the frequency shaping of T :

- **crossover** f_c : the frequency where $|T| = 1$ (0 dB);
- **phase margin** $\text{PM} = 180^\circ + \angle T(j2\pi f_c)$ — distance from oscillation;
- **gain margin** $\text{GM} = -|T|_{\text{dB}}$ where $\angle T = -180^\circ$;
- **sensitivity peak** $S_{\text{pk}} = \max|1/(1+T)|$ — a single robustness number ($S_{\text{pk}} < 1.5$: well damped; $S_{\text{pk}} > 2$: pronounced ringing).

Because several parameters are uncertain in reality, every design is evaluated over **corners**, not just nominal values: the TIP125 gain spread ($\beta_0 = 500 \dots 4000$), its frequency derating at low bias current, and (for the V loop) the load impedance and the output-capacitor ESR.

2 Device data used in the models

Op-amps (U1–U4): gain-bandwidth product $\text{GBW} = 5.7 \text{ MHz}$, slew rate $2.8 \text{ V}/\mu\text{s}$. The open-loop gain is modelled as a dominant-pole integrator $A(s) = 2\pi \cdot \text{GBW} / s$ and enters the compensator exactly through the noise-gain form (Section 3.1). Slew rate is a large-signal limit and does not appear in the AC model.

TIP125 PNP Darlington (Q2/Q4): the datasheet's small-signal current gain (Fig. 7, PNP @ 3 A) is fitted by a two-pole roll-off,

$$\beta(s) = \frac{\beta_0}{(1 + s/\omega_{\beta 1})(1 + s/\omega_{\beta 2})}, \quad f_{\beta 1} \approx 60 \text{ kHz}, \quad f_{\beta 2} \approx 300 \text{ kHz},$$

which reproduces the curve well (Figure 2). **This roll-off — not the op-amp — is the dominant high-frequency dynamic of both loops.** Two spreads are swept as corners:

- β_0 : datasheet minimum 1000, typical 2000–4000 (DC gain, Fig. 9); 500 is added as cold/low-current pessimism. Loop gain scales 1:1 with β_0 , so the crossover slides almost a decade across production spread.

- Low-bias derating: Fig. 7 is specified at 3 A. At small collector current f_T drops, so a derated corner with $f_{\beta 1,2} = 20/100$ kHz is checked (“LB” in the tables).

Minimum-load current sink (≈ 12 mA). The production hardware places a constant-current sink on each output (BCP56 Q304, diode bias D1006, $47\ \Omega$ degeneration R320), connected so that it **bypasses the current shunt**. The pass transistor therefore never idles below ≈ 12 mA — without it, only the 1 mA sense-divider current flows at no load, and nothing bounds how far f_β collapses. Small-signal the sink is high-impedance ($\approx 100\ \text{k}\Omega$), so it does not change any load case; its benefit is **bias**: the LB corner becomes a **guaranteed pessimistic floor** instead of an open-ended assumption, and if 12 mA is enough to hold the Fig.-7 poles, the LB corner does not apply at all. The tables below report the worst case both ways. Because the sink bypasses the shunt, the current loop sees no 12 mA offset — CC regulates the true load current and setpoints below 12 mA remain usable; the pass device also never cuts off at no load, and the sink provides 12 mA of down-programming current.

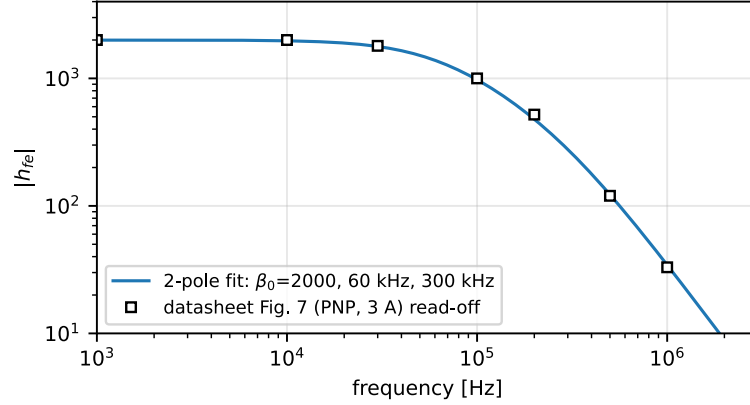


Figure 2: Two-pole fit of the TIP125 small-signal current gain vs. datasheet Fig. 7 read-off points (PNP, $I_C = 3$ A, $V_{CE} = 4$ V).

Diodes D1/D2 (bias steering into the driver base) appear only as the divider $R_3/(R_3 + r_d) \approx 0.97$ with $r_d \approx 26\ \Omega$.

3 Voltage loop (sheet V_loop, p. 2)

3.1 Stage 1 — error amplifier / compensator U1

V_{set} enters the (+) input; the feedback voltage V_{fb} enters the (-) input through R_1 , with the feedback impedance Z_f from output to (-):

$$Z_f(s) = \left(R_2 + \frac{1}{sC_1} \right) \parallel \frac{1}{sC_3}, \quad H_1(s) = -\frac{Z_f/R_1}{1 + \frac{1+Z_f/R_1}{A(s)}}.$$

This is a **type-II compensator**: an integrator $1/(sR_1(C_1 + C_3))$, a zero at $f_z = 1/(2\pi R_2 C_1)$ and a high-frequency pole at $f_p = (C_1 + C_3)/(2\pi R_2 C_1 C_3)$. With the optimized values $R_1 = 10\ \text{k}$, $R_2 = 18\ \text{k}$, $C_1 = 47\ \text{n}$, $C_3 = 22\ \text{p}$:

$$f_z \approx 188\ \text{Hz}, \quad \text{mid-band gain } \frac{R_2}{R_1} = 1.8\ (5.1\ \text{dB}), \quad f_p \approx 400\ \text{kHz}.$$

The finite GBW adds an effective pole near $\text{GBW}/(1 + R_2/R_1) \approx 2\ \text{MHz}$ — a minor extra lag, but included exactly.

3.2 Stage 2 — driver and pass transistor (Q1, R3, R4, D1, Q2)

U1's output drives Q1 (common emitter, degeneration R_4) through D1; Q1's collector current is the base current of the TIP125, giving a transconductance

$$H_2(s) = \frac{i_{c2}}{V_e} = \underbrace{\frac{R_3}{R_3 + r_d}}_{\approx 0.97} \cdot \underbrace{\frac{1}{R_4 + r_{e1}}}_{g_{m1}} \cdot \beta_2(s) \quad [\text{A/V}], \quad G_m = \frac{0.97 \cdot \beta_0}{150\ \Omega} = 13\ \text{A/V} \quad (\text{at } \beta_0 = 2000).$$

3.3 Stage 3 — output network (R5, C2, load R6)

Q2's collector acts as a current source into the output capacitor branch and the load:

$$H_3(s) = Z_{\text{out}}(s) = \left(R_5 + \frac{1}{sC_2} \right) \parallel \left[R_{\text{load}} \parallel \left(R_{\text{esrL}} + \frac{1}{sC_{\text{load}}} \right) \right].$$

The load “R6” is **whatever the user connects**, so Z_{out} – and with it the whole loop – changes with the load: a resistive load leaves the $1/(\omega C_2)$ slope up to the ESR zero, while a large load capacitor drags the crossover down by decades. This is the defining difficulty of the voltage loop.

3.4 Stage 4 – sense amplifier U2

A resistive difference amplifier (plus the op-amp’s GBW pole at $\text{GBW}/(1 + R_9/R_{11}) \approx 5.3 \text{ MHz}$):

$$H_4 = \left(1 + \frac{R_9}{R_{11}} \right) \frac{R_7}{R_7 + R_{10}} \cdot \frac{1}{1 + s/\omega_{U2}} = \frac{1}{15} \cdot \frac{1}{1 + s/\omega_{U2}},$$

so the loop regulates $V_{\text{out}} = 15 V_{\text{set}}$.

3.5 Loop gain and compensator design

Figure 3 shows the assembled loop at the nominal load ($100 \Omega \parallel 1 \mu\text{F}$): the integrator crosses the flat mid-band on a clean single slope, the zero at 188 Hz has long since returned the compensator phase, and the crossover lands at $f_c \approx 22 \text{ kHz}$ with $\text{PM} \approx 76^\circ$.

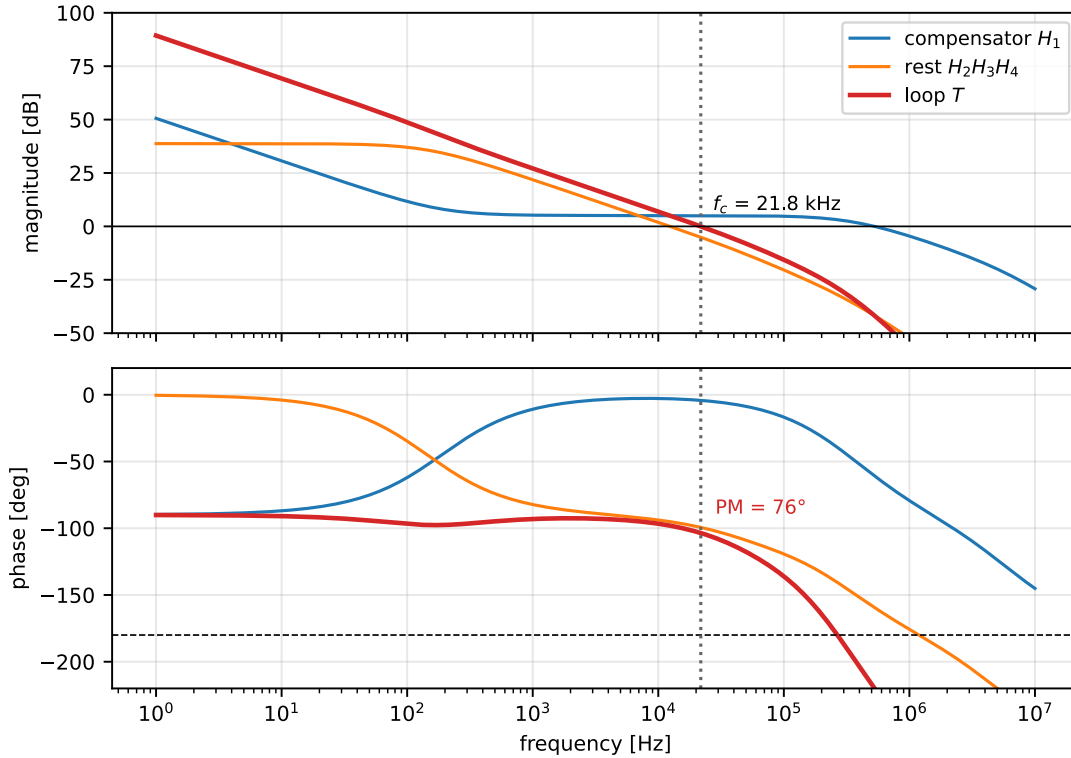


Figure 3: V loop at $\beta_0 = 2000$, load $100 \Omega \parallel 1 \mu\text{F}$: compensator H_1 , rest $H_2H_3H_4$, and loop gain T .

The values were found by a grid search over E12 components with the corner requirements $\text{PM} \geq 45^\circ$ ($\geq 35^\circ$ for the extreme $1000 \mu\text{F}$ case), $\text{GM} \geq 10 \text{ dB}$ and $S_{\text{pk}} \leq 1.7$ over **all** corners (β spread $\times$ ESR $\times$ load set), maximising the worst-case crossover. Two rules of thumb summarise why the result looks the way it does:

1. **The zero must sit below the biggest-cap crossover.** A large load capacitor drags f_c down; if f_c falls below f_z the loop reverts to integrator-on-top-of-cap-pole (-180° slope) and the margin collapses. $f_z = 188 \text{ Hz}$ keeps even the $\beta = 500 + 1000 \mu\text{F}$ corner ($f_c \approx 116 \text{ Hz}$) inside the zero’s phase lead.
2. **The mid-band gain sets the light-load crossover.** $R_2/R_1 = 1.8$ keeps the crossover at $\beta_0 = 4000$ ($f_c \approx 45 \text{ kHz}$) clear of the β poles even when the design is denied the help of C_2 ’s ESR zero (ESR corner $50 \text{ m}\Omega$).

3.6 Robustness across loads and corners

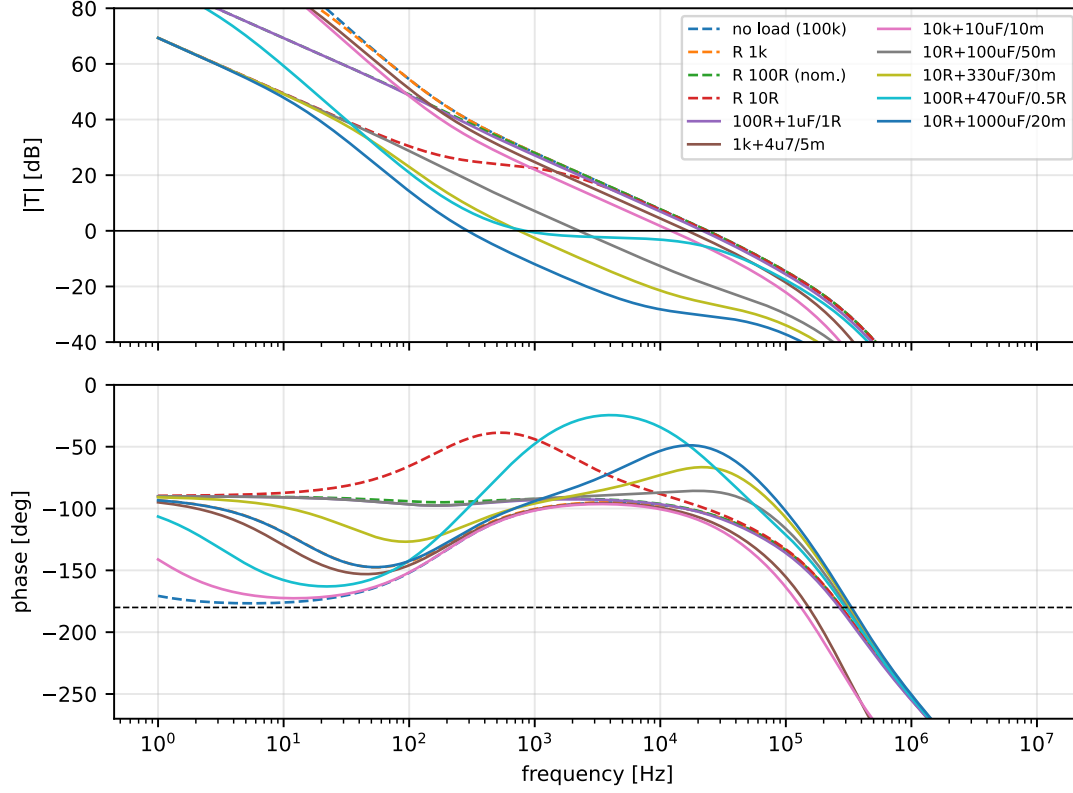


Figure 4: V-loop gain across 11 load types at $\beta_0 = 2000$ (dashed: purely resistive). Big capacitors move the crossover down by up to two decades — the compensator must be stable along the whole trajectory.

load case	f_c [Hz]	PM [°]	GM [dB]	S_{pk}	worst PM [°]	floor PM [°]	floor S_{pk}
no load (100 k)	23 930	75.6	28.1	1.15	47.8	37.0	1.99
1 k Ω	23 926	75.7	28.1	1.15	47.8	37.0	1.99
100 Ω (nominal)	23 885	76.1	28.1	1.15	48.0	37.5	1.97
10 Ω (heavy)	23 440	79.7	28.3	1.14	50.1	42.2	1.83
100 Ω + 1 μ F/1 Ω	21 801	76.4	28.9	1.15	50.9	40.2	1.87
1 k + 4.7 μ F/5 m Ω	16 534	75.8	24.3	1.17	55.7	46.1	1.69
10 k + 10 μ F/10 m Ω	12 240	77.6	26.3	1.15	62.4	53.6	1.51
10 Ω + 100 μ F/50 m Ω	2 252	89.7	44.2	1.01	86.1	84.5	1.05
10 Ω + 330 μ F/30 m Ω	749	81.1	48.7	1.01	62.7	62.7	1.03
100 Ω + 470 μ F/0.5 Ω	838	126.1	32.2	1.07	61.0	61.0	1.26
10 Ω + 1000 μ F/20 m Ω	292	61.9	52.2	1.01	40.0	40.0	1.48

Table 1: V loop per load: left half at the typical corner ($\beta_0 = 2000$, $f_\beta = 60/300$ kHz, ESR 0.2 Ω). **worst PM** = worst case over the β /ESR corners with datasheet β poles (valid if the 12 mA sink holds the Fig.-7 poles); **floor PM** / **floor S_{pk}** = additionally including the ± 3 low-bias pole corner — the guaranteed bound at ≥ 12 mA bias. The global worst with datasheet poles is the 1000 μ F case (40.0°, $\beta = 4000$) — a load property, not a bias property.

β corner	f_c range [Hz]	worst PM [°]	min GM [dB]	max S_{pk}
$\beta = 500$	116 ... 6 188	40.0	35.0	1.48
$\beta = 1000$ (min spec)	178 ... 12 279	49.5	28.9	1.21
$\beta = 2000$ (typical)	292 ... 23 930	61.9	22.9	1.33
$\beta = 4000$ (typ. peak)	522 ... 44 766	47.8	16.9	1.64
$\beta = 1000$, low-bias	178 ... 10 920	49.1	19.6	1.49
$\beta = 2000$, low-bias	291 ... 18 403	37.0	13.6	1.99

Table 2: V loop worst case per β corner, over all 11 loads and both ESR values. The binding corners are $\beta = 4000$ with low-ESR C2 (fast side) and the doubly derated low-bias case (slow side); both remain stable with sensible margin. The two low-bias rows are the floor guaranteed by the 12 mA minimum-load sink – they no longer bind if 12 mA holds the datasheet β poles.

3.7 Setpoint step and prefilter

The closed-loop response from the setpoint is

$$\frac{V_{out}}{V_{set}} = \frac{G_m(s) Z_{out} (1 + Z_f/R_1)}{1 + T},$$

whose numerator carries the **noise-gain** term $(1 + Z_f/R_1)$ – the compensator zero appears in the reference path with a zero at

$$f_{z,ref} = \frac{1}{2\pi(R_1 + R_2)C_1} \approx 121 \text{ Hz}.$$

A setpoint edge therefore feeds forward and overshoots by $\approx 54\%$ – **not** ringing (the loop is critically damped); it disappears if the setpoint never contains energy above that zero. The clean fix is an RC low-pass in front of the (+) input with its pole **matched to the zero**: $\tau = (R_1 + R_2)C_1 \approx 1.32 \text{ ms}$, e.g. **27 k + 47 n**. The zero then cancels the filter pole (Figure 5). Note that $R_1 C_1$ alone ($470 \mu\text{s}$) is not the match – it leaves about 13 %.

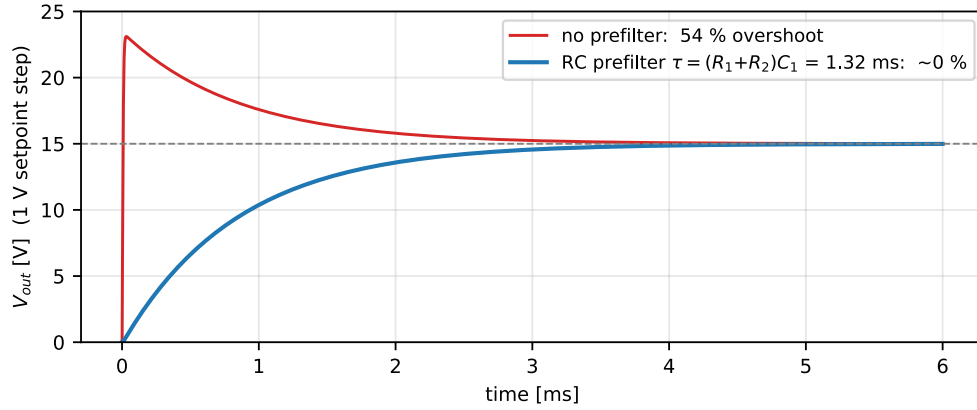


Figure 5: Closed-loop 1 V setpoint step ($\beta_0 = 2000$, nominal load) with and without the matched RC prefilter.

4 Current loop (sheet I_loop, p. 3)

4.1 Stages 1, 2 and 4

The error amplifier U3 and the driver Q3/Q4 are identical in topology to the voltage loop ($R_8 = 10 \text{ k}$ input resistor, $R_{15} = 150 \Omega$ degeneration, same TIP125 model). The optimized compensator is a **pure integrator** – R_{12} and C_5 stay unpopulated:

$$H_1(s) = -\frac{1/(sR_8C_4)}{1 + (1 + 1/(sR_8C_4))/A(s)}, \quad C_4 = 27 \text{ n} \rightarrow \text{unity gain at } \frac{1}{2\pi R_8 C_4} \approx 589 \text{ Hz}.$$

The sense amplifier U4 is non-inverting with gain $1 + R_{16}/R_{17} = 12$; its noise gain of 12 pulls the op-amp pole down to $\text{GBW}/12 \approx 475 \text{ kHz}$ – low enough to matter alongside the β poles:

$$H_4(s) = \frac{12}{1 + s/(2\pi \cdot 475 \text{ kHz})}, \quad V_{fb} = 12 \cdot R_{21} \cdot I = 2.4 \text{ V/A} \quad (I \approx 417 \text{ mA per volt of } V_{set}).$$

4.2 Stage 3 – the flat, load-independent plant

Here the schematic differs decisively from the voltage loop: the output-capacitor branch ($R_{19} + C_6$) **and** the load R_{20} both return to the **top of the shunt** R_{21} , not to ground. Every milliampere that Q4 sources must therefore pass through the shunt, and Kirchhoff's current law gives, exactly and at every frequency,

$$i_{R_{21}} = i_{C_6} + i_{load} = i_{c4} \Rightarrow H_3(s) = \frac{V_{sh}}{i_{c4}} = R_{21} = 0.2 \Omega.$$

The load impedance, C_6 and R_{19} **drop out of the loop gain completely** – they only set the voltage at the output node, which the regulated variable never sees. The loop is identical for a short, a resistor, or 1000 μF . Including Q4's finite output resistance ($r_o \approx 2 \text{ k}$ pessimistic) and output capacitance ($C_{ob} \leq 300 \text{ pF}$) merely adds a static gain error up to $\approx 5 \%$ at light load (a current-divider effect, calibration territory) and **no dynamics** below the MHz range.

4.3 Loop gain – why a pure integrator

With a flat plant, the “rest” of the loop is a constant $0.97\beta_0 R_{21} \cdot 12/R_{15} \approx 31$ (30 dB at $\beta_0 = 2000$) with the $\beta/\text{U4}$ poles hanging off the top (Figure 6). A single integrator crosses it on a clean -20 dB/dec slope: crossover

$$f_c \approx \frac{0.97\beta_0 R_{21}(1 + R_{16}/R_{17})}{2\pi R_8 C_4 R_{15}} = 17.6 \text{ kHz at } \beta_0 = 2000.$$

A voltage-loop-style zero ($R_{12} > 0$) would be actively harmful here: it flattens the compensator above f_z , and with the 8:1 β -gain spread the crossover would slide across that flat region straight into the β poles. **Keep $R_{12} = 0$.** $C_4 = 27 \text{ n}$ is the speed/margin knee: 22 n is $\approx 20 \%$ faster but sits at the acceptance edge (low-bias $S_{pk} \approx 2$), 47 n is the conservative fallback ($\text{PM} \geq 65^\circ$).

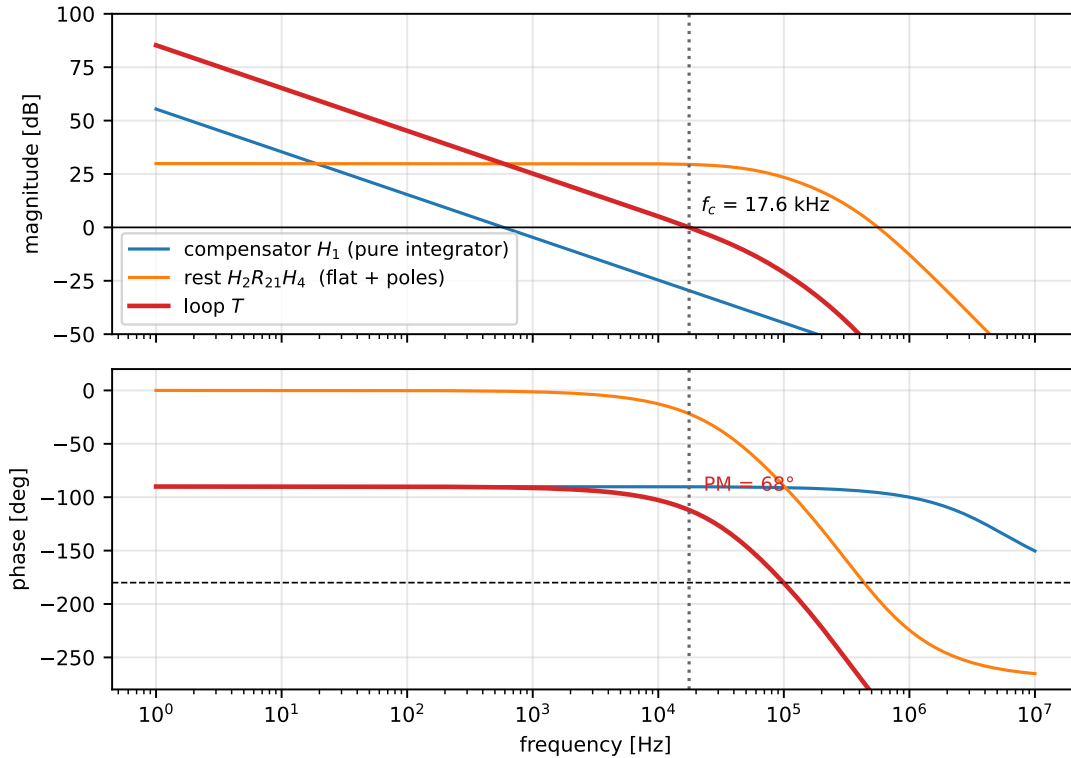


Figure 6: I loop at $\beta_0 = 2000$: integrator, flat rest, and loop gain. The phase budget at crossover is -90° (integrator) minus the β -pole lag.

4.4 Corner sweep

β corner	f_c [Hz]	PM [°]	GM [dB]	S_{pk}
$\beta = 500$	4 582	84.2	33.1	1.08
$\beta = 1000$ (min spec)	9 082	78.5	27.1	1.15
$\beta = 2000$ (typical)	17 597	68.0	21.0	1.30
$\beta = 4000$ (typ. peak)	32 148	51.5	15.0	1.60
$\beta = 1000$, low-bias	8 437	61.2	20.3	1.39
$\beta = 2000$, low-bias	14 662	43.5	14.3	1.78

Table 3: I loop margins per β corner. The numbers are **identical for every load** – the load does not appear in $T(s)$ (Section 4.2). Worst case with datasheet β poles: 51.5° ($\beta = 4000$); the low-bias rows are the 12 mA-sink-guaranteed floor (43.5° worst). The sink bypasses the shunt, so it adds **no offset** to the sensed current – CC regulates the true load current and setpoints below 12 mA remain usable.

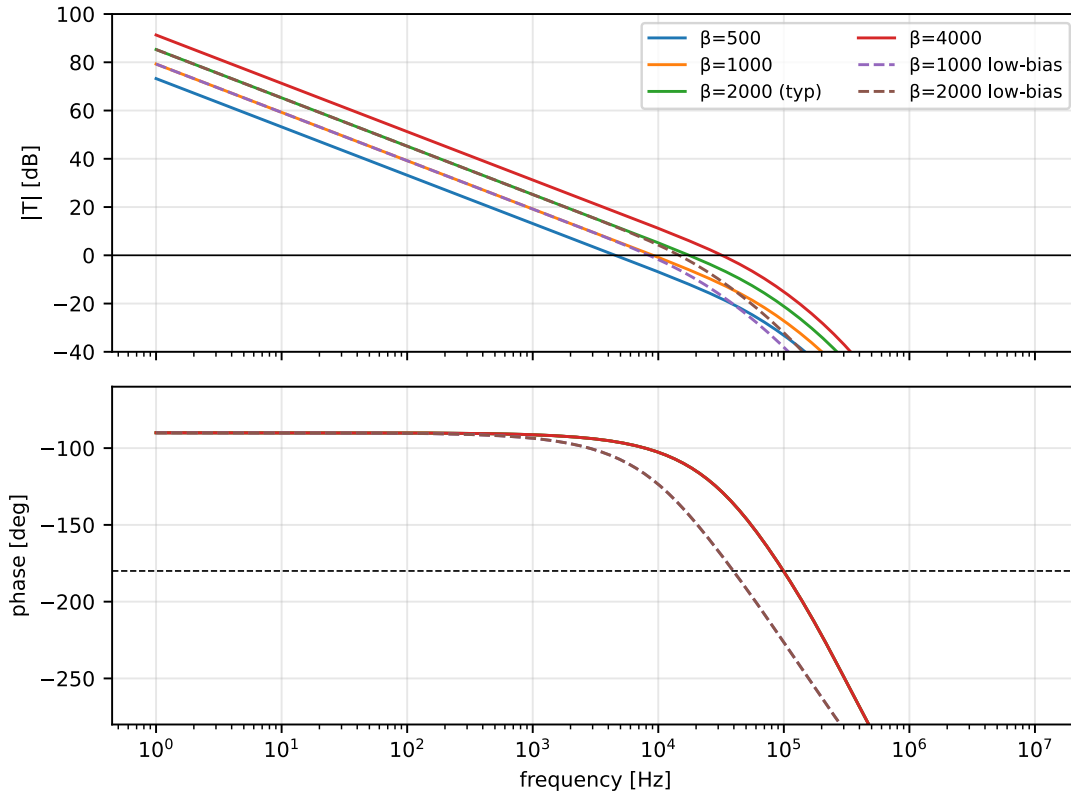


Figure 7: I-loop gain across the TIP125 β corners (dashed: low-bias derated poles). Only the gain scales; the pole structure – and therefore the phase curve – barely moves.

4.5 Setpoint step – the feed-forward spike

Because the plant has **no inertia** (a flat 0.2Ω , no output pole in the current path), a V_{set} edge feeds through the noise-gain term $(1 + Z_f/R_8) \rightarrow 1$ directly to the driver and commands $\Delta I \approx G_m \Delta V_{set} \approx 13 \text{ A/V}$ for the few microseconds ($1/2\pi f_c \approx 9 \mu\text{s}$) the integrator needs to take over – the small-signal model shows a spike of more than 2000 % (Figure 8, left). Hardware clips it (op-amp swing and $2.8 \text{ V}/\mu\text{s}$ slew, Q3's emitter resistor), but the lesson stands: **the current setpoint must never step hard**. The matched RC prefilter (pole on the reference zero at $1/2\pi R_8 C_4 = 589 \text{ Hz}$, i.e. $\tau = R_8 C_4 = 270 \mu\text{s}$, **10 k + 27 n**) removes the spike while keeping the setpoint response fast, because the compensator zero cancels the filter pole (Figure 8, right).

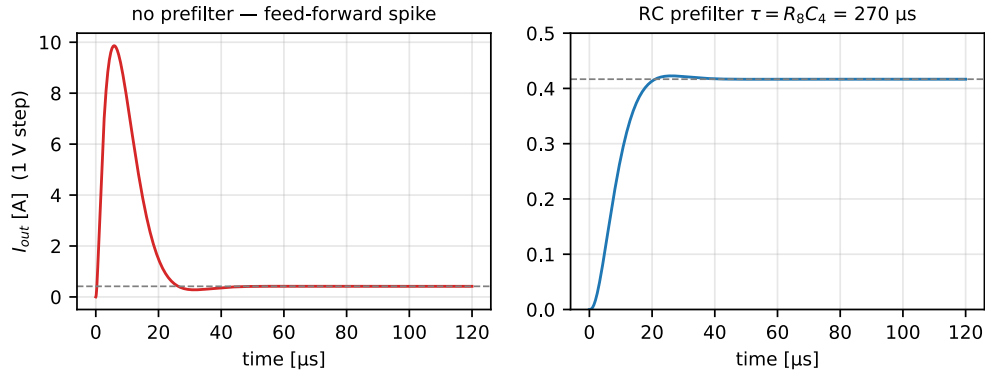


Figure 8: I loop, 1 V setpoint step at $\beta_0 = 2000$. Left: unfiltered — the feed-forward spike reaches amperes for a few μs (model; hardware clips it). Right: with the matched 270 μs prefilter — $\approx 1.4\%$ residual, settled in $\approx 20\ \mu\text{s}$.

5 Setpoint prefilters — general rule

Both overshoots have the same root cause: the compensator's feedback impedance appears in the **reference path** as $(1 + Z_f/R_{\text{in}})$, whose zero lets a setpoint edge feed forward. The matched RC prefilter puts its pole exactly on that zero:

$$\tau_{\text{filt}} = (R_{\text{in}} + R_{\text{zero}}) \cdot C_{\text{int}}$$

loop	reference zero	matched τ	suggested RC	result (1 V step)
V loop	$1/2\pi(R_1 + R_2)C_1 \approx 121\ \text{Hz}$	1.32 ms	27 k + 47 n (or 10 k + 120 n)	54 % $\rightarrow \approx 0\%$
I loop	$1/2\pi R_8 C_4 \approx 589\ \text{Hz}$	270 μs	10 k + 27 n	2267 % $\rightarrow 1.4\%$, settle $\approx 20\ \mu\text{s}$

Table 4: Matched setpoint prefilters. The filter is a series R + C-to-ground directly at the error amp's (+) input (high impedance). A firmware DAC ramp of $\approx 1\ \text{ms}$ is the part-free alternative. Note the values track the compensator: change C_1/C_4 and the filter caps follow.

Because the reference zero contains only compensator parts, the cancellation is independent of the TIP125 β and of the load.

6 Summary

part	schematic first guess	final value	role
V: R_1	10 k	10 k (keep)	input / gain reference
V: R_2	10 k	18 k	zero at 188 Hz, mid-band gain 1.8
V: C_1	1 n	47 n	integrator; zero with R_2
V: C_3	—	22 p	HF pole $\approx 400\ \text{kHz}$
I: R_8	10 k	10 k (keep)	input / gain reference
I: C_4	—	27 n	pure integrator, unity at 589 Hz
I: R_{12}, C_5	—	DNP	a zero would be harmful (flat plant)
V prefilter	—	27 k + 47 n	kills 54 % setpoint overshoot
I prefilter	—	10 k + 27 n	kills the feed-forward current spike

Table 5: Final component values (all E12).

Resulting worst-case margins (with the $\approx 12\ \text{mA}$ minimum-load sink keeping the pass device biased) — V loop: $\text{PM} \geq 40^\circ$ with datasheet β poles (the binding case is the extreme $10\ \Omega + 1000\ \mu\text{F}$ load at $\beta = 4000$; every normal load stays $\geq 47.8^\circ$), $\text{GM} \geq 16.9\ \text{dB}$; the $\div 3$ -derated low-bias floor bottoms at 37° . Typical- β margins $62\text{--}126^\circ$. I loop: $\text{PM} \geq 51.5^\circ$, $\text{GM} \geq 15\ \text{dB}$ over the full β spread (low-bias floor 43.5°), for **any** load. Before the sink, the low-bias corner was an open-ended assumption — the pass device idled at the $1\ \text{mA}$ divider current and nothing bounded the f_β collapse; now it is a guaranteed floor.

Not covered by these AC models, to be verified on hardware: the CV $\leftrightarrow$ CC handover through the D1/D2 diode-OR (the losing loop's integrator saturates and must slew back — an anti-windup clamp shortens the transition),

how far the TIP125 β poles actually derate at the 12 mA minimum bias (bounded by the LB corner, but Fig. 7 is specified at 3 A), and thermal drift of the TIP125 parameters.